

Forensics Investigation

Exercise: 1.5 | Memory Dump Analysis

Objectives:

The primary objective is to conduct a thorough investigation into a suspected memory dump with the goal of identifying the source, scope, and impact of the attack.

Incident Details:

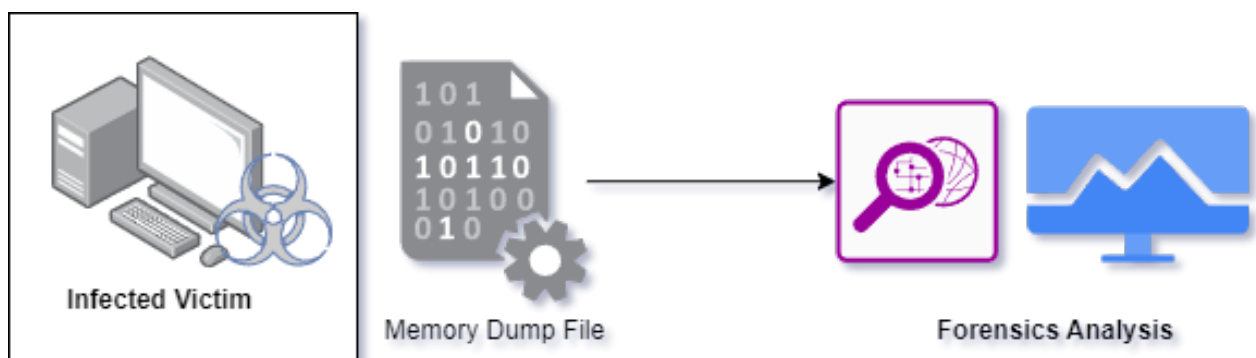
Investigate the suspicious endpoint compromise activity by the Endpoint monitoring systems. Investigate the .RAW file to identify the targeted IP, source, and the root cause of the activity.

Memory Dump

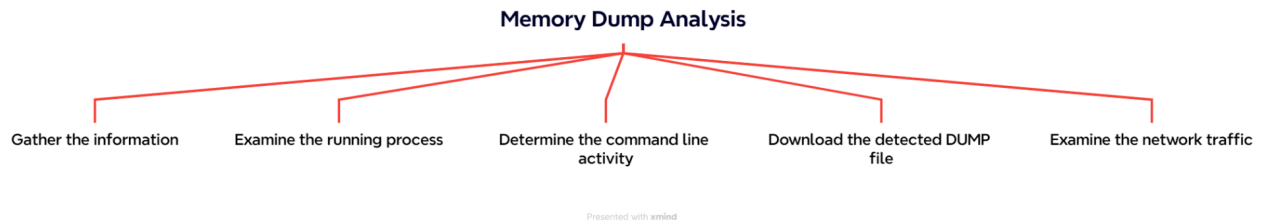
suspected.raw

About Memory Dump Analysis:

Memory dump analysis plays a critical aspect of identifying the root cause and incident response. A memory dump, also known as a core dump or a crash dump, is a file that captures the contents of a computer's volatile memory at a specific point in time. This snapshot of memory can provide valuable information for understanding system behaviour, identifying anomalies, and investigating security incidents.



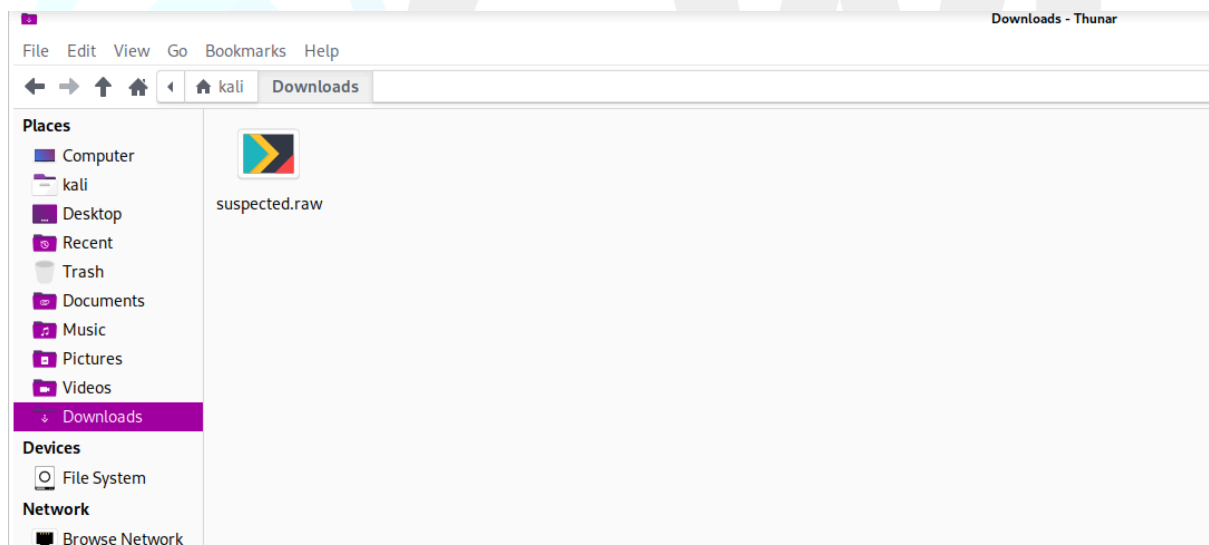
Investigation Mindmap:



Step:01

The first step of the investigation begins by analysing the given memory dump. We have provided you with the suspected RAW file which has been extracted from one of the compromised host machines.

Download the given memory dump on your own host machine in an isolated environment.



For our investigation we will be processed using the **volatility3-2.4.1**, feel free to download from the below mentioned link

Unset

<https://github.com/volatilityfoundation/volatility3/archive/refs/tags/v2.4.1.zip>

```
(kali@kali)-[~/Desktop/volatility3-2.4.1]
└─$ sudo python3 vol.py -f /home/kali/Downloads/suspected.raw windows.info
Info
Volatility 3 Framework 2.4.1
Progress: 100.00 Development doc PDB scanning finished README.md requirements-minimal.txt setup.py volatility3
Variable Value LICENSE.txt sypy.ini requirements-dev.txt requirements.txt test volatility3.egg-info
Kernel Base 0xf80105200000 volatility3-2.4.1
DTB 0x1aa000
Symbols file:///home/kali/Desktop/volatility3-2.4.1/volatility3/symbols/windows/ntkrnlmp.pdb/769C521E4833ECF72E21F02BF33691A5-1.json.xz
Is64Bit True
IsPAE False
layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf80105e0f368
Major/Minor 15.19041
MachineType 34404
KeNumberProcessors 4
SystemTime 2024-02-20 14:03:22
NtSystemRoot C:\Windows
NtProductType NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Tue Oct 11 07:04:26 1977
```

As per the above result the file shell.exe seems to be a suspicious process running.

Determine the command line activity

After determine the suspicious process its recommend to cross verify the process with the command line activity execute the below mentioned commands and observe the findings

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```
sudo python3 vol.py -f <File_Path>/suspected.raw windows.cmdline.CmdLine
```

```
3348 RuntimeBroker.exe C:\Windows\System32\RuntimeBroker.exe -Embedding
6108 ApplicationFrameHost.exe C:\Windows\System32\ApplicationFrameHost.exe -Embedding
4112 svchost.exe C:\Windows\System32\svchost.exe -k LocalService -p -s BthAvctpSvc
6788 SkypeApp.exe "C:\Program Files\WindowsApps\Microsoft.SkypeApp_14.53.77.0_x-ww_kzfqkx38z5c\SkyApp.exe" -ServerName:App.AppXFn3yxqvgaw9fpmhy98Fr3y01dt15b.mca
7892 RuntimeBroker.exe C:\Windows\System32\RuntimeBroker.exe -Embedding
7655 svchost.exe C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p -s lmhosts
8108 svchost.exe C:\Windows\System32\svchost.exe -k LocalService -p -s fdPHost
8108 svchost.exe C:\Windows\System32\svchost.exe -k LocalServiceNetwork -p -s NcdAutoSetup
5424 svchost.exe C:\Windows\System32\svchost.exe -k LocalServiceAndNoImpersonation -p -s FDRestub
4332 svchost.exe C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p -s DeviceAssociationService
2908 dashost.exe (f37eae1e-bbe3-4c33-8e97313447daefae)
7352 svchost.exe C:\Windows\System32\svchost.exe -k LocalServiceAndNoImpersonation -p -s SSDPSRV
7640 powershell.exe "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
5524 conhost.exe C:\Windows\System32\conhost.exe 0x4
1572 svchost.exe C:\Windows\System32\svchost.exe -k LocalServiceAndNoImpersonation -p -s wcnscvc
8172 svchost.exe C:\Windows\System32\svchost.exe -k LocalService -p -s WbioSrv
7396 shell.exe "C:\Users\Emp10\Downloads\shell.exe"
4316 cmd.exe cmd
3680 conhost.exe \?C:\Windows\System32\conhost.exe 0x4
1364 msedge.exe "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --no-startup-window
3508 msedge.exe "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=crashpad-handler --user-data-dir=C:\Users\Emp10\AppData\Local\Microsoft\Edge\User Data --prefetch:4 --monitor-self-annotation-ptype=crashpad-
handler --database=C:\Users\Emp10\AppData\Local\Microsoft\Edge\User Data\Crashpad --annotation=IsofficialBuild-1 --annotation=channel- --annotation=chromium-version=121.0.6167.184 --annotation=exe=C:\Program Files (x86)\Microsoft\Edg
e\Application\msedge.exe --annotation=plat-win64 --annotation=prod-microsoft edge --annotation=ver=121.0.2277.128 --initial-client-data=0x234,0x238,0x23c,0x238,0x244,0x7ff2c2a5b0f9,0x7ff2c2a5b0f9,0x7ff2c2a5b0f9
5192 msedge.exe "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=utility --utility-sub-type=network.mojm.NetworkService --lang=en-US --service-sandbox-type=none --no-appcompat-clear --mojo-platform-channel-
handle=2268 --field-trial-handle=1936,1,5981183711536507931,3275571256574668311,262144 --variations-seed-version /prefetch:3
6544 msedge.exe "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=utility --utility-sub-type=storage.mojm.StorageService --lang=en-US --service-sandbox-type=service --no-appcompat-clear --mojo-platform-chann
el-handle=2496 --field-trial-handle=1936,1,5981183711536507931,3275571256574668311,262144 --variations-seed-version /prefetch:8
6156 msedge.exe "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=renderer --instant-process --no-appcompat-clear --first-renderer-process --lang=en-US --js-flags --ms-user-locale= --device-scale-factor=1 --n
um-raster-threads=2 --enable-main-frame-before-activation --renderer-client-id=14 --time-ticks-at-unix-epoch=1708436570368627 --launch-time-ticks=713038381 --mojo-platform-channel-handle=3268 --field-trial-handle=1936,1,598118371153650
7931,3275571256574668311,262144 --variations-seed-version /prefetch:1
7316 msedge.exe "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=renderer --no-appcompat-clear --lang=en-US --js-flags --ms-user-locale= --device-scale-factor=1 --num-raster-threads=2 --enable-main-frame-bef
ore-activation --renderer-client-id=17 --time-ticks-at-unix-epoch=1708436570368627 --launch-time-ticks=713327381 --mojo-platform-channel-handle=5332 --field-trial-handle=1936,1,5981183711536507931,3275571256574668311,262144 --variation
s-seed-version /prefetch:1
6308 MsMpEng.exe "C:\ProgramData\Microsoft\Windows Defender\Platform\4.18.23110.3-0\MsMpEng.exe"
4348 NLSrv.exe "C:\ProgramData\Microsoft\Windows Defender\Platform\4.18.23110.3-0\NLSrv.exe"
3184 svchost.exe C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p -s WDISystemHost
4808 SystemSettings.exe "C:\Windows\ImmersiveControlPanel\SystemSettings.exe" -ServerName:microsoft.windows.immersivecontrolpanel
1440 audiodg.exe C:\Windows\System32\AUDIOIOG.EXE 0x36c
4456 MRVC120.exe "C:\Users\Emp10\Downloads\MRVC120.exe"
5848 SearchProtocol "C:\Windows\System32\SearchProtocolHost.exe" Global\VsGthrCtrlPipeMsGthrPipe5_Global\VsGthrCtrlPipeMsGthrPipe5 1-2147483646 "Software\Microsoft\Windows Search" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT
; MS Search 4.0 Robot)"
2428 SearchFilterHo "C:\Windows\System32\SearchFilterHost.exe" 0 800 804 812 8192 808 784
```

As per the above result we did not observe any suspicious command execution associated with the executables.

Generally many malicious exe are been associated with multiple command execution activity, it's been recommended to cross-verify the activity which are associated with the detected exe

Download the detected DUMP file

Next step of the investigation is to download the detected suspicious file for further investigation to, using our raw image file we can easily retrieve the file using the PID value associate with it, execute the below mentioned commands and observed the result

Unset

```
sudo python3 vol.py -f <file_path>/suspected.raw windows.pslist --pid 7396 --dump
```

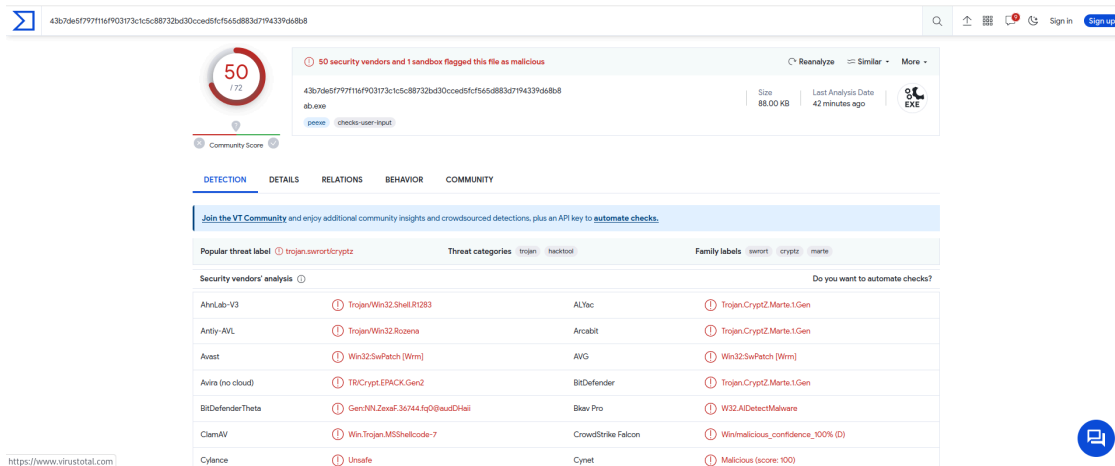
```
(kali@kali)-[~/Desktop/volatility3-2.4.1]
$ sudo python3 vol.py -f /home/kali/Downloads/suspected.raw windows.pslist --pid 7396 --dump
[sudo] password for kali:
Volatility 3 Framework 2.4.1
Progress: 100.00
PDB scanning finished
PDB Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output
7396 5716 shell.exe 0xa90ac08ac080 1 - 1 True 2024-02-20 13:54:28.000000 N/A pid.7396.0x400000.dmp
```

After extracting the dump the next step is to check for the strings associated with the dump file

```
(kali@kali)-[~/Desktop/volatility3-2.4.1]
$ strings pid.7396.0x400000.dmp
!This program cannot be run in DOS mode.
Rich
Computer
.text
.rdata
.data
.rsrc
.PAG
h`8@
S@H@I
7C/K
A/I7
speY
'aQ
!M"*
h`l;
jRoj
;$u
D$$[[aYZQ
]h32
hws2_ThLw&
TPh)
PPPP@P@Ph
```

In majority of the cases we might find interesting value but we did not observed any interesting value in our case it recommended to analyse the dump via virustotal, upload the extracted dump directly into the virustotal for further analysis

Navigate to Virustotal : <https://www.virustotal.com/gui/>



50 security vendors and 1 sandbox flagged this file as malicious

43b7de5f797f116f903173c1c5c88732bd30cced5f565d883d7194339d68b8

Size: 88.00 KB | Last Analysis Date: 42 minutes ago

Popular threat label: trojan.sswortcryptz

Threat categories: trojan, hacktool

Family labels: smoot, cryptz, marte

Security vendors' analysis:

Vendor	Detection	Category	Score
AhnLab-V3	Trojan.Win32.Shell.R283	ALYac	100
Antiy-AVL	Trojan.Win32.Rczena	Arcabit	100
Avast	Win32:Swpatch [Wrm]	AVG	100
Avira (no cloud)	TRICrypt.EP&CK.Gen2	BitDefender	100
BitDefender Theta	Gen:NLZexaF.3c744.fg0@baidChai	Bkav Pro	100
ClamAV	Win.Trojan.MSShellcode-7	CrowdStrike Falcon	100
Cylance	Unsafe	Cynet	100

Do you want to automate checks?

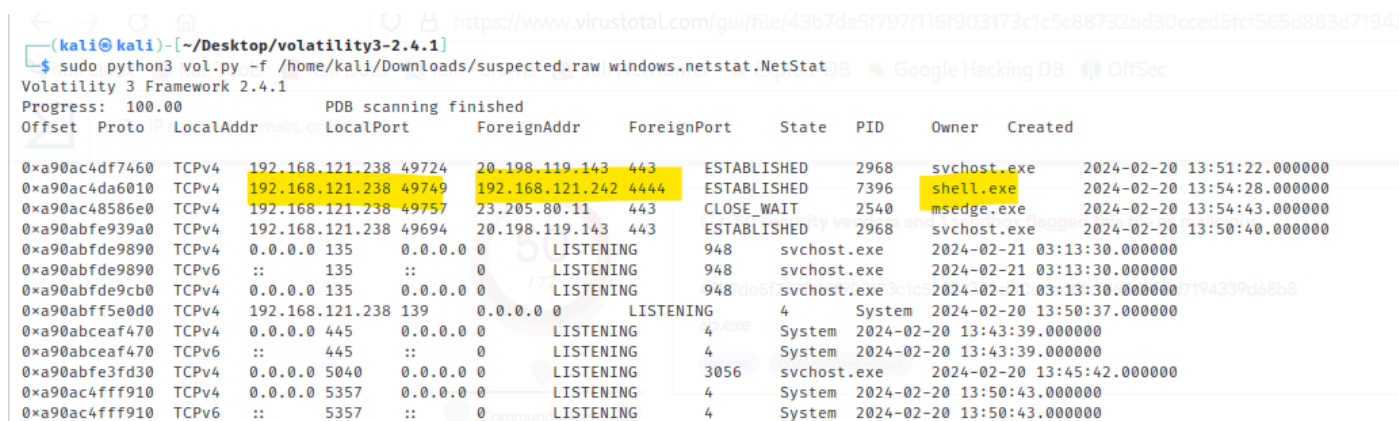
As per the result from the VT, we confirm that the detected file is malicious.

Examine the network traffic

After determining it suspiciousness its recommend to determine whether is there any network activity associated with the detected file, to list all network connection execute the below mentioned command and observed the result

Unset

```
sudo python3 vol.py -f <file_path>/suspected.raw windows.netstat.NetStat
```



```
(kali@kali)~[/Desktop/volatility3-2.4.1]
$ sudo python3 vol.py -f /home/kali/Downloads/suspected.raw windows.netstat.NetStat
Volatility 3 Framework 2.4.1
Progress: 100.00
PDB scanning finished
```

Offset	Proto	IP	LocalAddr	LocalPort	ForeignAddr	ForeignPort	State	PID	Owner	Created
0xa90ac4df7460	TCPv4		192.168.121.238	49724	20.198.119.143	443	ESTABLISHED	2968	svchost.exe	2024-02-20 13:51:22.000000
0xa90ac4da6010	TCPv4		192.168.121.238	49749	192.168.121.242	4444	ESTABLISHED	7396	shell.exe	2024-02-20 13:54:28.000000
0xa90ac48586e0	TCPv4		192.168.121.238	49757	23.205.80.11	443	CLOSE_WAIT	2540	msedge.exe	2024-02-20 13:54:43.000000
0xa90abfe939a0	TCPv4		192.168.121.238	49694	20.198.119.143	443	ESTABLISHED	2968	svchost.exe	2024-02-20 13:50:40.000000
0xa90abfde9890	TCPv4		0.0.0.0	135	0.0.0.0	0	LISTENING	948	svchost.exe	2024-02-21 03:13:30.000000
0xa90abfde9890	TCPv6	::	::	::	::	::	LISTENING	948	svchost.exe	2024-02-21 03:13:30.000000
0xa90abfde9cb0	TCPv4		0.0.0.0	135	0.0.0.0	0	LISTENING	948	svchost.exe	2024-02-21 03:13:30.000000
0xa90abff5e0d0	TCPv4		192.168.121.238	139	0.0.0.0	0	LISTENING	4	System	2024-02-20 13:50:37.000000
0xa90abceaf470	TCPv4		0.0.0.0	445	0.0.0.0	0	LISTENING	4	System	2024-02-20 13:43:39.000000
0xa90abceaf470	TCPv6	::	::	::	::	::	LISTENING	4	System	2024-02-20 13:43:39.000000
0xa90abfe3fd30	TCPv4		0.0.0.0	5040	0.0.0.0	0	LISTENING	3056	svchost.exe	2024-02-20 13:45:42.000000
0xa90ac4fff910	TCPv4		0.0.0.0	5357	0.0.0.0	0	LISTENING	4	System	2024-02-20 13:50:43.000000
0xa90ac4fff910	TCPv6	::	::	::	::	::	LISTENING	4	System	2024-02-20 13:50:43.000000

As per the above result we clearly observed that the detected executable **shell.exe** has been successfully communicating to an IP **192.168.121.242** via port **4444**

Thank You!

Hope you have learnt a lot after practically completing
Blue Team Fundamentals Exercise.

If you like the investigation & the lab scenario,
feel free to tag us & share your achievement / progress
on social media platforms :)

For Enterprise Red Team / Blue Team / Purple Team
Cyber Range Labs / Training please email :
support@cyberwarfare.live

To know more visit at : <https://cyberwarfare.live>